Alcohols

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DATE PLACEHOLDER

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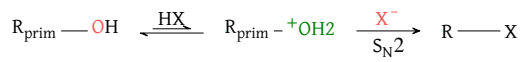
1 Alcohol Reactions

Deprotonating alcohol requires a stronger base than RO^-. In other words, the base must have a stronger affinity to the H^+ than the alkoxide (RO^-). If the alcohol is acidic, water will usually suffice in this matter. However, in cases where water is more acidic than the alcohol itself, H^+ can be donated to the alcohol and a subsequent H_2O L group will be formed, leaving the resulting product to have a highly reactive, empty atomic orbital on the relevant C after the H_2O L group departs.

This phenomenon presents a range of complications when dealing with alcohol reactions. For instance, let's say that we want to replace the $-OH$ with a different functional group like a halide X. This can occur either through S_N2 or S_N1.



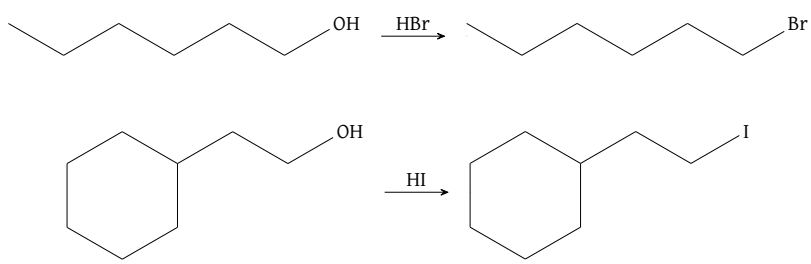


Figure 1: S_N2 of alcohols; needs strong H^+ with a good nucleophile HCl, HBr, HI

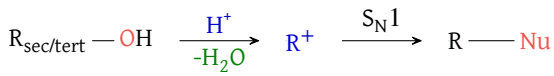


Figure 2: S_N1 of alcohols; *E1 may occur as well, leading to an alkane end product, but* S_N1 *dominates for* R_{sec} *in* H_2O

However, mixtures ensue due to several different mechanisms.

Core Concept 1.1: Hydride Shifts

Hydride shifts refer to the movement of a intramolecular hydrogen towards a neighboring empty p orbital.

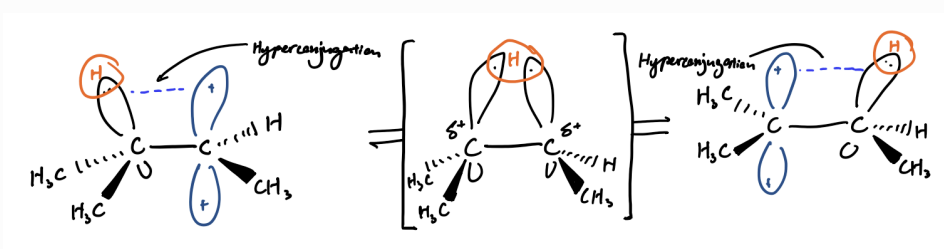


Figure 3: Hydride Shift Mechanism in SN1 Reaction

This is particularly an issue if the shift is favorable, such as the move from a R_{sec}^+ to a R_{tert}^+. Degenerate shifts ($R_{sec}^+ \rightarrow R_{sec}^+$ or $R_{tert}^+ \rightarrow R_{tert}^+$) are less common, but can still occur, resulting in the halide attacking at a unintended location.

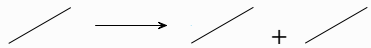


Figure 4: Filler caption

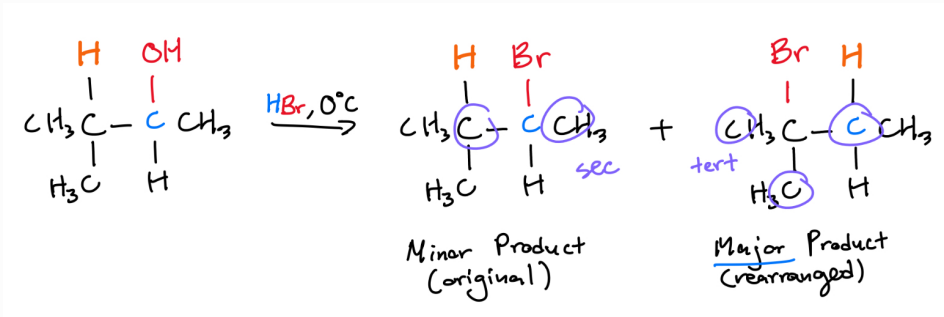
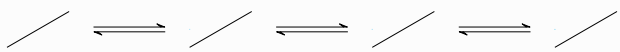


Figure 5: Hydride Shift Resulting in Regioisomeric Products

In specific cases like ring structures, these hydride shifts happen stereospecifically.



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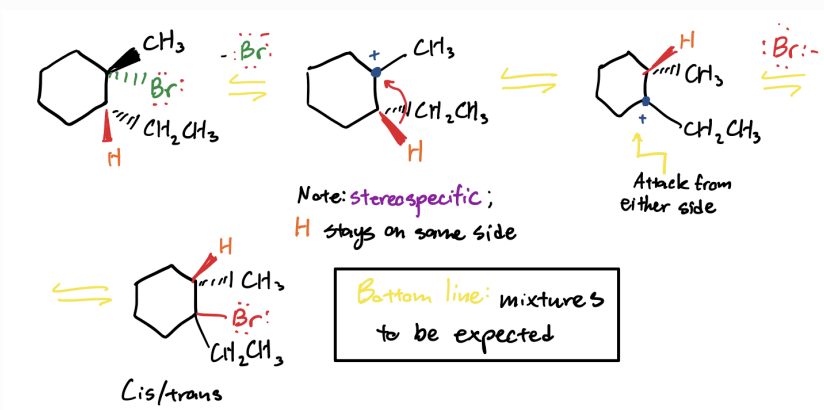


Figure 7: Stereospecific Hydride Shift in Ring Structures

Core Concept 1.2: E1 Reactions

E1 reactions become particularly more prevalent at higher temperatures and in the absence of a viable nucleophile. *Notice how the H^+ detachment at the last step happens at the tertiary position and not the secondary due to greater stablity from more hyperconjugation.*

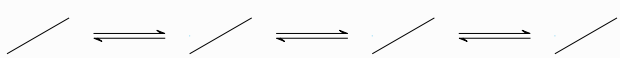


Figure 8: Filler caption

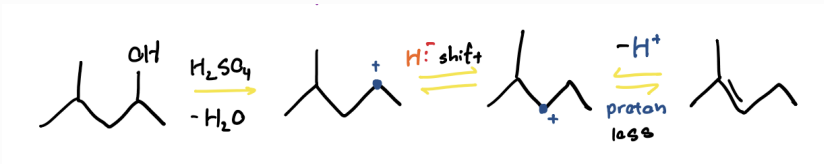
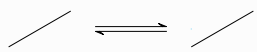


Figure 9: Elimination (E1) Mechanism in Alcohols

Core Concept 1.3: Alkyl Shifts

Additionally, entire alkyl or R shifts can occur, although not as commonly as hydride shifts because of the greater E_a. These shift can occur even in R_{prim} – OHs.



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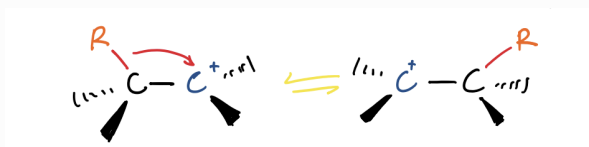


Figure 11: Alkyl Shift Mechanism in Alcohol Reactions

However, alkyl shifts are especially favorable and fast when relieving ring stain.

Figure 12: Filler caption

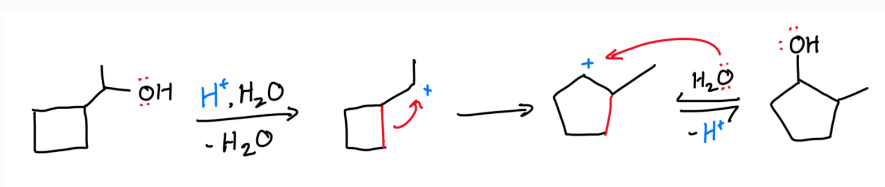
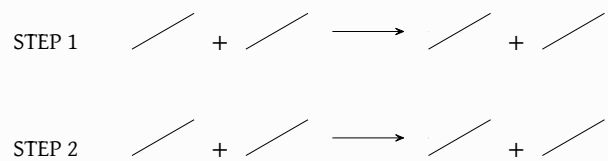


Figure 13: Ring Expansion via Alkyl Shift

So, how do we actually turn an ROH into RNu without all of these potential issues that are bound to happen? The solution is to use inorganic derivatives as reactive or isolable intermediates.

Core Concept 1.4: Substituting OH with Br

Chemists use PBr_3 to replace OH groups with Br: $3\,ROH + PBr_3 \longrightarrow 3\,RBr + H_3PO_3$. The mechanism for this reaction is as follows.



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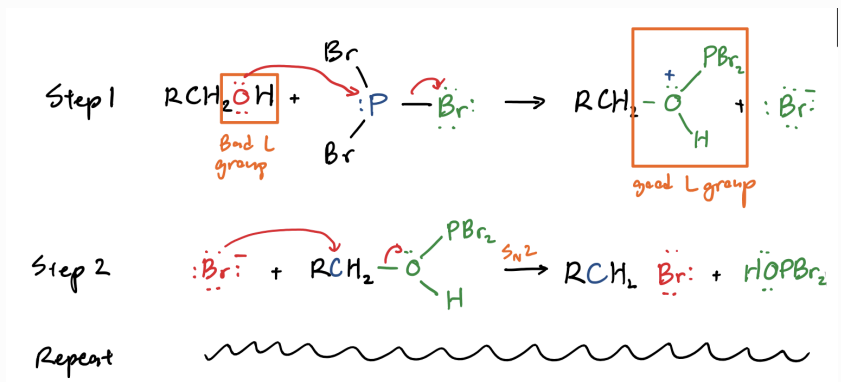


Figure 15: Mechanism of Alcohol Bromination using PBr3

Core Concept 1.5: Substituting OH with Cl

Chemists use $SOCl_2$ to replace OH groups with Cl: ROH + $SOCl_2$ $\longrightarrow$ RCl + SO_2 + HCl. The mechanism for this reaction is as follows.

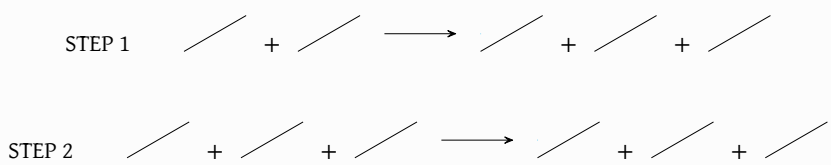


Figure 16: Filler caption

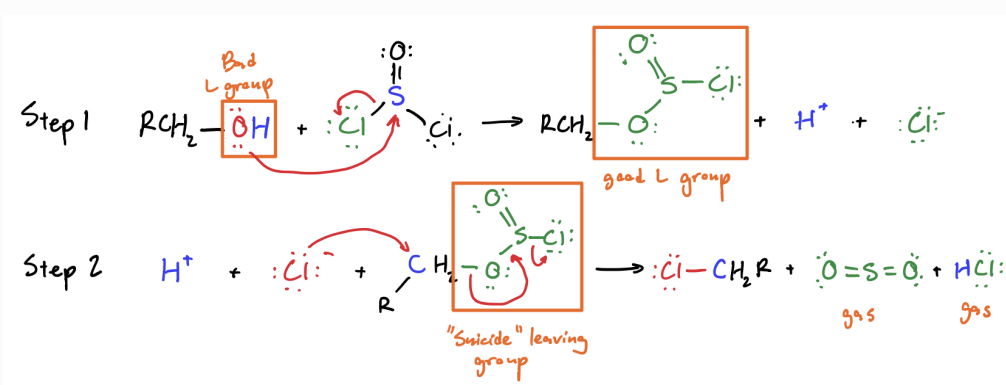


Figure 17: Mechanism of Alcohol Chlorination using SOCl2

It is important to note here that pyradine is used as a solvent in this reaction in order to neutralize the HCl

byproduct from further protonating the alcohol or the product, effectively preventing unintended compounds from forming.

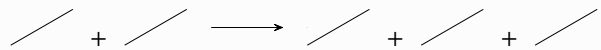


Figure 18: filler caption

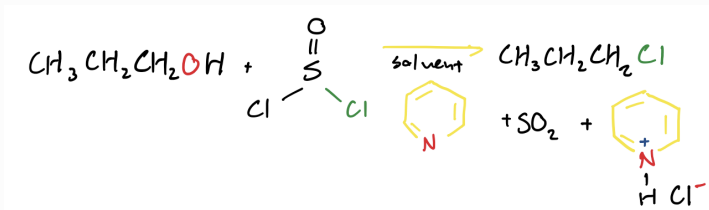
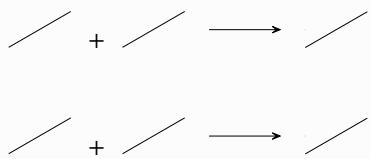


Figure 19: Reaction of Alcohol with Thionyl Chloride (SOCl2)

Core Concept 1.6: Substituting OH with Good Leaving Groups

To substitute OH with a good leaving group, chemists use isolable sulfonates such as the following molecules and cooresponding reactions.



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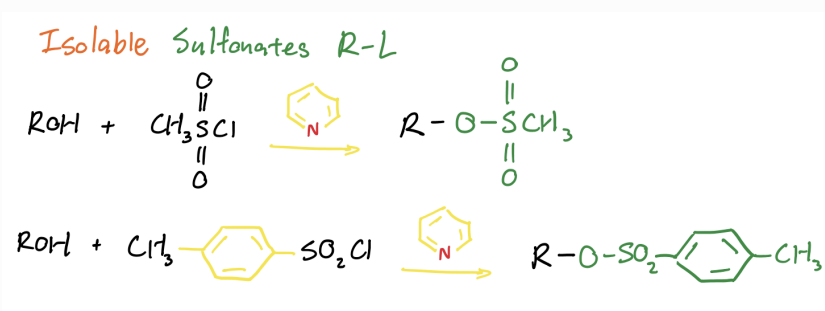

Figure 21: Formation of Isolable Sulfonate Esters

Notice that the displaced Cl^- after the sulfonates attach to the R group is stablized as a floating ion via solvation by the pyradine solvent as seen in the reaction above with $SOCl_2$.